

Tracking Flu and RSV Hospital Admissions in the United States: A Live Dashboard

Abstract

This project built a live, publicly accessible dashboard that tracks weekly flu and RSV hospital admissions for every US state. It uses official CDC hospital data from the National Healthcare Safety Network, retrieved automatically through the Delphi Epidata API, and adjusts counts for state population. Analysis of the 2024–25 through early 2026–27 seasons found that the 2024–25 flu season was more than twice as severe as 2025–26 in California. The 2025–26 season also spread across the country from east to west. The latest data, from September 2026, suggests the 2026–27 flu season may be starting early. The dashboard is built in Python with Streamlit and Plotly and updates itself as new weekly data is released.

Introduction

Influenza (flu) and respiratory syncytial virus (RSV) are two of the most common causes of respiratory hospitalization in the United States. Both follow a seasonal pattern, and both put the heaviest strain on hospitals during winter. RSV is especially serious for infants and older adults, while flu affects all age groups.

The burden is large. CDC estimates that flu has caused between 120,000 and 710,000 hospitalizations per season since 2010 [7], and that RSV caused an estimated 190,000 to 350,000 hospitalizations during the 2024–25 season [8]. CDC also classified 2024–25 as a high-severity flu season, with the highest cumulative flu hospitalization rate recorded since 2010–11 [9]. This context matters for the findings below, which show 2024–25 as the worse of the two seasons studied.

Because these diseases rise and fall quickly, timely data matters. Knowing where cases are rising, and how the current season compares with past ones, helps hospitals plan staffing and beds and helps public health officials time vaccination campaigns and warnings. Much of this data is public, but it is spread across technical databases that are hard for non-specialists to use.

This project addresses that gap. Its goal was to build an automatically updating dashboard that turns official CDC hospital data into clear, state-by-state views of flu and RSV activity. The project asked four research questions:

1. How do flu and RSV hospital admissions change over the course of a year?
2. How does the severity of one season compare with another?
3. Do flu and RSV peak at the same time, or at different times?
4. Does the flu season reach all states at once, or does it spread geographically?

Data sources

The project combines two public datasets.

Data	Source	What it provides	Update frequency
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Data	Source	What it provides	Update frequency
Hospital admissions	CDC National Healthcare Safety Network (NHSN), accessed through the Delphi Epidata API at Carnegie Mellon University	Weekly counts of new hospital admissions with laboratory-confirmed flu or RSV, for each US state	Weekly, about 6–7 days after each week ends
Population	US Census Bureau 2024 state population estimates	Resident population of each state, DC, and Puerto Rico	Yearly

NHSN collects admission counts directly from hospitals. Reporting became mandatory in November 2024, so data from that point on covers most US hospitals. The Delphi Epidata API makes this data available in a consistent format, which allows the dashboard to download it automatically. The two signals used were `confirmed_admissions_flu_ew` and `confirmed_admissions_rsv_ew`, covering October 2024 to the present.

Methods

The project moved through four stages, from exploring the data in a notebook to publishing a live dashboard.

The four stages were: collecting data from the Delphi Epidata API, processing it, building the dashboard with Streamlit and Plotly, and deploying it through GitHub and Streamlit Community CloudData collection

Data was first explored in a Jupyter Notebook, using Python 3.11 through Anaconda. Weekly flu and RSV admissions for all states were requested from the Delphi Epidata API with the `requests` library and loaded into tables with `pandas`. A backup copy was saved as a CSV file, so the dashboard still works if the API is unavailable.

Data processing

- **Dates:** CDC reports data by epidemiological week (for example, week 202606). Each week was converted to the Saturday it ends on, using the `epiweeks` library.
- **Population adjustment:** Weekly admissions were divided by state population and multiplied by 100,000, giving admissions per 100,000 residents. This allows fair comparison between large and small states.
- **Seasons:** Each respiratory season was defined as August 1 to July 31, so a winter season is labeled by both years (for example, 2025–26). Each week was also numbered by weeks since August 1, so different seasons can be lined up on one chart.
- **Rising states:** A state was counted as rising if admissions grew by more than 10% from the previous week. To reduce noise from very small numbers, only states with at least 5 admissions in the previous week were included.

Dashboard design

The dashboard was built with Streamlit, a Python library for web apps, and Plotly for interactive charts. It includes:

- **Controls:** choice of disease (flu or RSV), measure (per 100k or total), and state.
- **Headline numbers:** national admissions for the latest week, the change from the previous week, the number of rising states, and the current season's peak so far.
- **State map:** admissions by state for any selected week.
- **Trend charts:** flu and RSV compared for one state, and each season overlaid on the same timeline.
- **Rising-states table:** the ten states with the largest weekly increase.

Data is cached for one hour, so the dashboard loads quickly while still picking up new weekly releases.

Deployment

The code was uploaded to a public GitHub repository and deployed on Streamlit Community Cloud, which gives the dashboard a public web address <https://flu-rsv-tracker.streamlit.app/>

The online version downloads fresh data from the API each time the cache expires, so it updates without manual work.

Findings

Flu and RSV hospital admissions peak every winter, and the 2024–25 flu season was more than twice as severe as 2025–26. Numbers below are read from the dashboard's charts and are approximate; hover over each chart to confirm exact values before submitting. Admissions are weekly confirmed hospital admissions from CDC's National Healthcare Safety Network.

1. Flu and RSV follow a strong winter seasonal pattern

In California, flu and RSV admissions stayed close to zero from spring through early fall. Both rose sharply each winter and peaked between late December and February. This pattern repeated in the 2024–25 and 2025–26 seasons, so hospitals face their greatest respiratory pressure in winter.

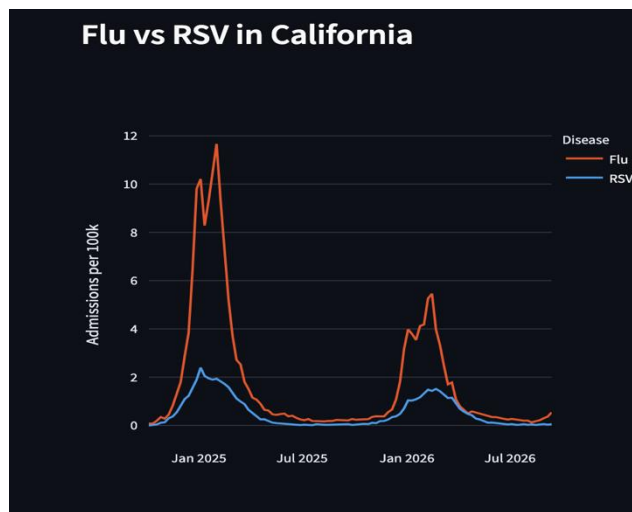


Figure 1. Weekly flu and RSV hospital admissions per 100,000 residents in California, October 2024 to September 2026

2. The 2024–25 flu season was more than twice as severe as 2025–26

Season	California peak week	Peak weekly admissions	Peak per 100k residents
2025–26	February 2026	about 2,150	about 5.5
2024–25	Early February 2025	about 4,600	about 11.7

The two seasons rose at similar times but reached very different heights. The dashboard does not show why. Possible explanations include the circulating flu strains, vaccine effectiveness, or population immunity, which would need further data to test.

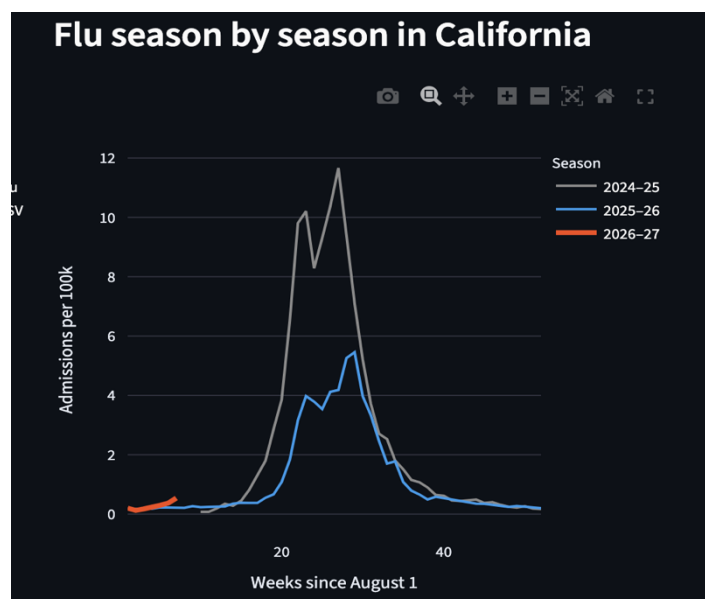


Figure 2. Weekly flu hospital admissions per 100,000 residents in California by respiratory season, measured in weeks since August 1

3. Flu causes far more admissions than RSV, and their timing varies

At the 2024–25 peak, California's flu admissions were roughly four to five times higher than RSV admissions. The order of the peaks changed between seasons:

- **2024–25:** RSV peaked in early January 2025, about four weeks before flu.
- **2025–26:** Both peaked at roughly the same time, in February 2026.

When the peaks overlap, hospitals handle both diseases at once, which may add strain on capacity.

4. The 2025–26 flu season spread from east to west

The national peak of the 2025–26 season came in the week ending January 3, 2026. That week's map shows the highest rates in the Northeast, Mid-Atlantic, and upper Midwest.

- **Most affected:** Pennsylvania had the highest rate, about 25 admissions per 100k. New York, Rhode Island, North Carolina, Minnesota, and Wisconsin were also high.
- **Least affected:** California, Oregon, Washington, Nevada, and Arizona had much lower rates that week.

California did not reach its own peak until about a month later, in February. This suggests the season moved across the country from east to west rather than hitting every state at once.

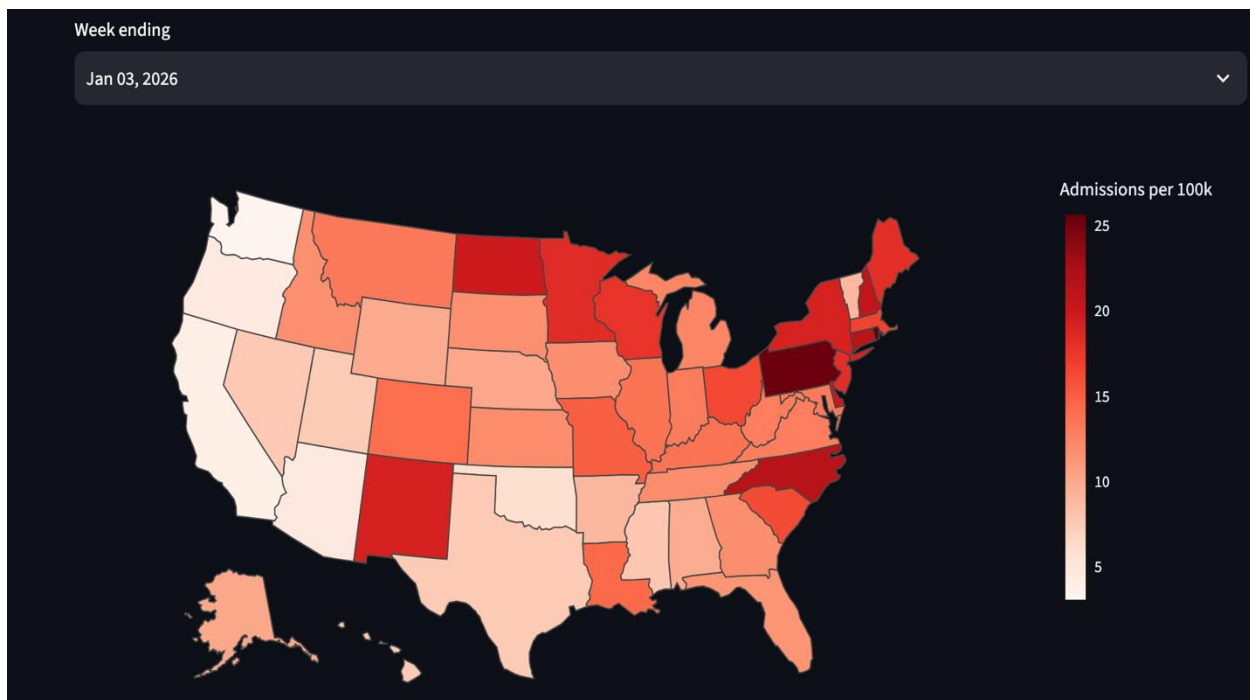


Figure 3. Flu hospital admissions per 100,000 residents by state, week ending January 3, 2026 (national peak of the 2025–26 season)

5. Adjusting for population changes which states look most affected

Raw counts are dominated by large states such as California and Texas. After converting to rates per 100k residents, smaller states such as Pennsylvania, Rhode Island, and New Mexico stood out during the January 2026 peak. Per-capita rates are therefore essential for comparing states fairly.

6. Early signs suggest the 2026–27 season may be starting early

In the week ending September 12, 2026, there were 1,479 flu admissions nationally, up 20.3% from the week before. 22 states rose by more than 10%. In the season chart, the 2026–27 line already sits slightly above where the previous two seasons were at the same point.

These are early numbers. The latest week is often revised as late reports arrive, and many fast-rising states started from small counts: Oklahoma rose from 7 to 18 admissions. Several more weeks of data are needed before concluding the season started early.



Figure 4. Dashboard summary for the week ending September 12, 2026: national flu admissions, week-over-week change, number of rising states, and the 2026–27 season peak so far

Fastest-rising states this week			
Only states with at least 5 admissions last week, so tiny numbers don't dominate.			
State	This week	Last week	Change (%)
Oklahoma	18	7	+157.1%
Indiana	15	7	+114.3%
New Jersey	40	19	+110.5%
Missouri	24	12	+100.0%
Kentucky	18	9	+100.0%
South Carolina	11	6	+83.3%
Virginia	28	16	+75.0%
Utah	19	11	+72.7%
Massachusetts	8	5	+60.0%
Minnesota	15	10	+50.0%

Figure 5. States with the largest week-over-week increase in flu admissions, week ending September 12, 2026 (states with at least 5 admissions the previous week)

7. Small-population states produce more volatile rates

On the September 12, 2026 map, Alaska had the highest per-100k rate in the country despite few admissions. With only about 740,000 residents, a handful of extra hospitalizations changes Alaska's rate sharply. Per-capita rates for small states should be read with caution, especially outside peak season.

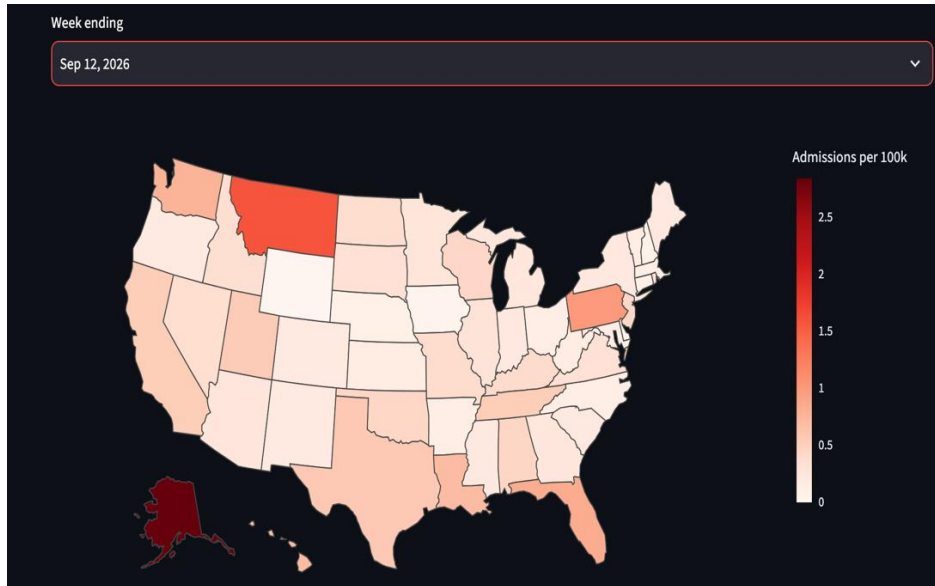


Figure 6. Flu hospital admissions per 100,000 residents by state, week ending September 12, 2026

Source for all figures: CDC National Healthcare Safety Network, via the Delphi Epidata API; population from US Census Bureau 2024 estimates.

Limitations

The findings should be read with several limitations in mind.

- **Reporting lag and revisions.** Each week's data arrives about 6–7 days after the weekends. The most recent week is often revised upward as late hospital reports arrive, so the latest numbers are the least certain.
- **Incomplete early data.** Hospital reporting was voluntary from May to October 2024, so data from that period covers fewer hospitals. RSV counts from before November 2024 are especially undercounted, which limited season comparisons to 2024–25 onward.
- **Only two full seasons.** With two complete seasons, the project can describe differences between them but cannot say what a "typical" season looks like.
- **Confirmed cases only.** The data counts only admissions with a positive laboratory test. Testing practices vary between hospitals and states, and the data does not capture people who were sick but not hospitalized.
- **Small-population volatility.** In small states such as Alaska or Wyoming, a few extra admissions can cause large jumps in per-100k rates. Week-over-week percentage changes can also be noisy when counts are small.
- **Population figures.** Rates use a single set of 2024 population estimates for all weeks, which slightly misstates rates as populations change over time.

- **State-level view.** State totals hide differences within states, such as between cities and rural areas. US territories other than Puerto Rico were excluded because data was mostly missing.
- **Descriptive, not causal.** The dashboard shows patterns but cannot explain why they happen. For example, it cannot show why the 2024–25 season was more severe.

Conclusion and future work

This project showed that public CDC data can be turned into a live, easy-to-use tracker of flu and RSV hospital admissions. The dashboard answered each research question:

1. **Seasonal pattern:** both diseases stay low from spring to fall and peak between late December and February.
2. **Season severity:** California's 2024–25 flu season peaked at more than twice the level of 2025–26.
3. **Timing:** flu causes far more admissions than RSV, and their peaks came four weeks apart in 2024–25 but overlapped in 2025–26.
4. **Geographic spread:** the 2025–26 season reached the Northeast and Midwest first and the West Coast about a month later.

The project also showed why adjusting for population matters, since per-100k rates change which states appear most affected. Early data from September 2026 suggests the next flu season may be starting ahead of schedule, which is exactly the kind of signal a live tracker is designed to catch.

Several extensions could make the tool more useful:

- **Forecasting:** predict admissions for the next two to four weeks using a time-series model.
- **Early-warning alerts:** flag states whose activity is unusually high compared with the same week in past seasons.
- **More diseases:** add COVID-19 admissions, which are available from the same NHSN source.
- **Wastewater data:** test whether wastewater virus levels rise before hospital admissions do, which could give earlier warning.
- **Finer detail:** show data by age group or by region within states, where available.

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